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EDUCATION

Ph.D in Probability and Mathematical Statistics

Academy of Mathematics and Systems Science (AMSS), Chinese Academy of Sciences

Sept, 2017 – Jun, 2022

Beijing, China

Bachelor in Statistics

Nankai University

Sept, 2013 – Jun, 2017

Tianjin, China

WORK EXPERIENCE

Assistant Professor

Department of Statistics and Data Science, Tsinghua University

Sept, 2026 – Present

Beijing, China

Postdoctoral Fellow

Department of Biostatistics, Harvard University

Sept, 2022 – Aug, 2026

Boston, United States

Advisor: Xihong Lin

RESEARCH AREAS

- Data Fusion
- Causal Identification and Inference
- Mendelian Randomization
- Large-scale Data Analysis
- Two-phase Sampling
- Statistical Learning Theory

PEER-REVIEWED PUBLICATIONS

(¹ Equal contribution; * Corresponding author)

1. **Wang, R.**, Wang Q.^{*}, and Miao, W. (2023), A robust fusion-extraction procedure with summary statistics in the presence of biased sources. *Biometrika*, 110, 1023–1040.
2. **Wang, R.**, Su, M., and Wang, Q.^{*} (2023), Distributed nonparametric imputation for missing response problems with massive data. *Journal of Machine Learning Research*, 68, 1–52.
3. Hu, W.¹, **Wang, R.**¹, Li, W.^{*}, and Miao, W.^{*} (2026), Semiparametric efficient fusion of individual data and summary statistics. *Journal of the American Statistical Association: T&M*, in press. arXiv:2210.00200.
4. **Wang, R.**, and Wang, Q.^{*} (2021), Determination and estimation of optimal quarantine duration for infectious diseases with application to data analysis of COVID-19. *Biometrics*, 78, 691–700.
5. Su, M. and **Wang, R.**^{*} (2026), Moment-assisted subsampling method for Cox proportional hazards model with large-scale data. *Journal of Computational and Graphical Statistics*, in press, arXiv:2501.06924.

6. **Wang, R.**, Wang, Q.^{*}, Miao, W., and Zhou, X. (2024), Sharp bounds for variance of treatment effect estimators in the finite population in the presence of covariates. *Statistica Sinica*, 34, 999–1021.
7. Su, M. and **Wang, R.**^{*} (2025), Subsampled one-step estimation for fast statistical inference. *Scandinavian Journal of Statistics*, 52, 2187–2208.
8. **Wang, R.**, Wang Q.^{*}, and Miao, W. (2025), A maximin optimal approach for sampling designs in two-phase studies. *Statistica Sinica*, in press. arXiv:2312.10596.
9. **Wang, R.**¹, Yi, M.¹, Chen, Z., and Zhu, S. (2022), Out-of-distribution generalization with causal invariant transformations. *IEEE Conference on Computer Vision and Pattern Recognition*, 375–385.
10. Yi, M.¹, **Wang, R.**¹, and Ma, Z. (2022), Characterization of excess risk for locally strongly convex population risk. *Advances in Neural Information Processing Systems* 36.
11. Yi, M., **Wang, R.**, Sun, J., Li, Z., and Ma, Z. (2023), Breaking correlation shift via conditional invariant regularizer. In *Proceedings of the 11th International Conference on Learning Representations*.
12. Yang, H., Liu, Z., **Wang, R.**, Lai, E., Schwartz, J., Baccarelli, A., Huang, Y. and Lin, X.^{*} (2025), Causal mediation analysis for integrating exposure, genomic, and phenotype data. *Annual Review of Statistics and Its Application*, 12, 337–360.
13. Su, M.¹, **Wang, R.**¹, and Wang, Q.^{*} (2022), A two-stage optimal subsampling estimation for missing data problems with large-scale data. *Computational Statistics and Data Analysis*, 173.
14. Wang, Q., Su, M.^{*}, and **Wang, R.** (2021), A beyond multiple robust approach for missing response problem. *Computational Statistics and Data Analysis*, 155.
15. Miao, W. ^{*}, Li, W., Hu, W., **Wang, R.**, and Geng, Z. (2021), Invited commentary: Estimation and bounds under data fusion. *American Journal of Epidemiology*, 191, 674–678.